In [1]:
```
import pandas as pd
```

In [2]:
```
import numpy as np
```

In [3]:
```
import matplotlib.pyplot as plt
```

Закрузка ФАЙЛА

In [6]:
```
df = pd.read_csv('/content/spotify_artist_streaming_2020_2025(2).csv')
```

Вывод всего датасета

In [7]:
```
df
```

Out[7]:

	track_id	track_name	artist_name	album_name	release_
0	346d85e2-ffde-41b3-854a-589ed880b91f	Storm	Ethereal	The Hollow Journeys	2021-
1	12bc268a-363b-4f19-8fa8-0ced7ae1e462	Silent	The Wolfs	Ancient Thunder	2024-
2	30d627d9-878d-4bb3-be59-71feb6ed4f3f	Shattered Lines	LolaOre	Twisted Harbors	2022-
3	85501bb9-d221-4339-9e2e-29b1b3daff0a	Breaking Roses	Clay Black	The Blinding Letters	2020-
4	7bf92bd2-c6fe-4869-a90e-242bb8bbe7a3	Lines	Knot Hart	The Cosmic Letters	2024-
...	...	...	...	...	
49995	38913908-e3ab-42a2-9a97-2d1847c21e3b	Flames	Luna Ink	The Shattered Tapes	2025-
49996	83653402-ca85-4583-b7ec-13ab3e82888e	Love	Knot Hart	Young II	2021-
49997	4398dd33-ecfc-4c1a-ab9e-248f38082318	The Waves	Jasper Jade	The Burning Journeys	2023-
49998	07c673e7-39a5-4c28-8789-163ae5f17f3f	Flying Eyes	ElaraRain	Frequencies	2020-
49999	888e233c-e7fd-4c0d-af8a-3cd98345a64e	Roads	Edge Brooks	Wicked III	2020-

50000 rows × 33 columns

первые 10 строчек

In [8]:
```
df.head(10)
```

Out[8]:

	track_id	track_name	artist_name	album_name	release_date
0	346d85e2-ffde-41b3-854a-589ed880b91f	Storm	Ethereal	The Hollow Journeys	2021-10-06
1	12bc268a-363b-4f19-8fa8-0ced7ae1e462	Silent	The Wolfs	Ancient Thunder	2024-10-26
2	30d627d9-878d-4bb3-be59-71feb6ed4f3f	Shattered Lines	LolaOre	Twisted Harbors	2022-11-12
3	85501bb9-d221-4339-9e2e-29b1b3daff0a	Breaking Roses	Clay Black	The Blinding Letters	2020-06-22
4	7bf92bd2-c6fe-4869-a90e-242bb8bbe7a3	Lines	Knot Hart	The Cosmic Letters	2024-01-22
5	e4a536ca-c6d4-4bad-97b5-d5e7b85c49e2	Wild Doors	Knot Hart	Jagged Gardens	2021-04-12
6	305b4f13-b2ac-47e9-a6f2-d0c15934f8b6	Skies	Knot Hart	The Ancient Tapes	2025-07-16
7	83c3a6be-6066-449a-a339-19afcde7cfb3	Wild	Blaze Project	Crystal	2023-12-12
8	437961fc-9ccd-4c86-8eb7-e96b6d2dc5c8	Neon Oceans	Clay Black	Broken V	2020-06-17
9	53df0c89-adc4-48b7-94e3-111e348cb17b	Eternal Angels	Phantasm	The Kind Notes	2020-06-09

10 rows × 33 columns

Кол-во строк и столбцов

In [9]:
```
df.shape
```

Out[9]:
```
(50000, 33)
```

Название всех столбцов

In [10…
```
df.columns
```

Out[10]:
```
Index(['track_id', 'track_name', 'artist_name', 'album_name', 'releas
e_date',
       'genre', 'duration_ms', 'popularity', 'danceability', 'energy',
'key',
       'loudness', 'mode', 'instrumentalness', 'tempo', 'stream_count',
       'country', 'explicit', 'label', 'release_year', 'release_month',
       'release_day_of_week', 'duration_minutes', 'popularity_categor
y',
       'loudness_category', 'key_name', 'mode_name', 'is_explicit_boo
l',
       'release_quarter', 'is_weekend_release', 'log_stream_count',
       'upbeat_score', 'artist_track_count'],
      dtype='object')
```

Типы данных столбцов

In [11…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 33 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   track_id             50000 non-null  object 
 1   track_name           50000 non-null  object 
 2   artist_name          50000 non-null  object 
 3   album_name           50000 non-null  object 
 4   release_date         50000 non-null  object 
 5   genre                50000 non-null  object 
 6   duration_ms          50000 non-null  int64  
 7   popularity           50000 non-null  int64  
 8   danceability         50000 non-null  float64
 9   energy               50000 non-null  float64
 10  key                  50000 non-null  int64  
 11  loudness             50000 non-null  float64
 12  mode                 50000 non-null  int64  
 13  instrumentalness     50000 non-null  float64
 14  tempo                50000 non-null  float64
 15  stream_count         50000 non-null  int64  
 16  country              50000 non-null  object 
 17  explicit             50000 non-null  bool   
 18  label                50000 non-null  object 
 19  release_year         50000 non-null  int64  
 20  release_month        50000 non-null  int64  
 21  release_day_of_week  50000 non-null  object 
 22  duration_minutes     50000 non-null  float64
 23  popularity_category  50000 non-null  object 
 24  loudness_category    50000 non-null  object 
 25  key_name             50000 non-null  object 
 26  mode_name            50000 non-null  object 
 27  is_explicit_bool     50000 non-null  bool   
 28  release_quarter      50000 non-null  object 
 29  is_weekend_release   50000 non-null  bool   
 30  log_stream_count     50000 non-null  float64
 31  upbeat_score         50000 non-null  float64
 32  artist_track_count   50000 non-null  int64  
dtypes: bool(3), float64(8), int64(8), object(14)
memory usage: 11.6+ MB
```

10 треков с максимальным количеством прослушиваний

In [12…
```
top10 = df.sort_values(by='stream_count', ascending=False).head(10)
```

Вывод нужных столбцов

In [13…
```
top10[['track_name', 'artist_name', 'genre', 'stream_count']]
```

Out[13]:

	track_name	artist_name	genre	stream_count
1528	The Roses	Grove Nova	Alternative	43345015
36998	Bitter	Flora Quinn	Rock	26635547
42813	The Streets	Ethereal	Pop	26314312
21643	Breaking Thunder	SethStone	Metal	20408155
49193	The Pages	IsleEyre	Pop	19265301
26622	Lights	Dean System	Hip-Hop	17238449
45015	Gentle	Rocco Ore	R&B	15776187
39012	Storm	Knot Hart	R&B	14545933
29091	Burning Islands	The Inks	Pop	14507867
13635	Broken Eyes	Prism Hart	R&B	14367106

треки после 2024

In [14…
```
df_after_2024 = df[df['release_year'] > 2024]
```

вывод кол-во найденных треков и первые строки

In [16…
```
print(f"Нашли треков после 2024 года: {len(df_after_2024)}")
df_after_2024[['track_name', 'artist_name', 'release_year']].head(10)
```

```
Нашли треков после 2024 года: 8318
```

Out[16]:

	track_name	artist_name	release_year
6	Skies	Knot Hart	2025
14	The Roads	Knot Hart	2025
17	Liquid Streets	ElaraRain	2025
18	Broken Ghosts	Sterling Zen	2025
24	The Tears	Sterling Zen	2025
27	Eternal	SethStone	2025
42	Waiting Night	Knot Hart	2025
46	Oceans	IsleEyre	2025
50	Empty Flames	Onyx Core	2025
60	Young Floors	Ethereal	2025

Среднее арифметическое и медиана

In [17…
```
print("Среднее popularity:", df['popularity'].mean())
```

```
Среднее popularity: 28.13682
```

In [18…
```
print("Медиана popularity:", df['popularity'].median())
```

```
Медиана popularity: 26.0
```

Минимум, максимум и стандартное отклонение для popularity

In [19…
```
print("Минимум popularity:", df['popularity'].min())
print("Максимум popularity:", df['popularity'].max())
print("Стандартное отклонение popularity:", df['popularity'].std())
```

```
Минимум popularity: 0
Максимум popularity: 92
Стандартное отклонение popularity: 15.999500616206952
```

наиболее часто встречающееся значение (моду):

In [21…
```
mode_genre = df['genre'].mode()
print("Мода для genre:")
print(mode_genre)
```

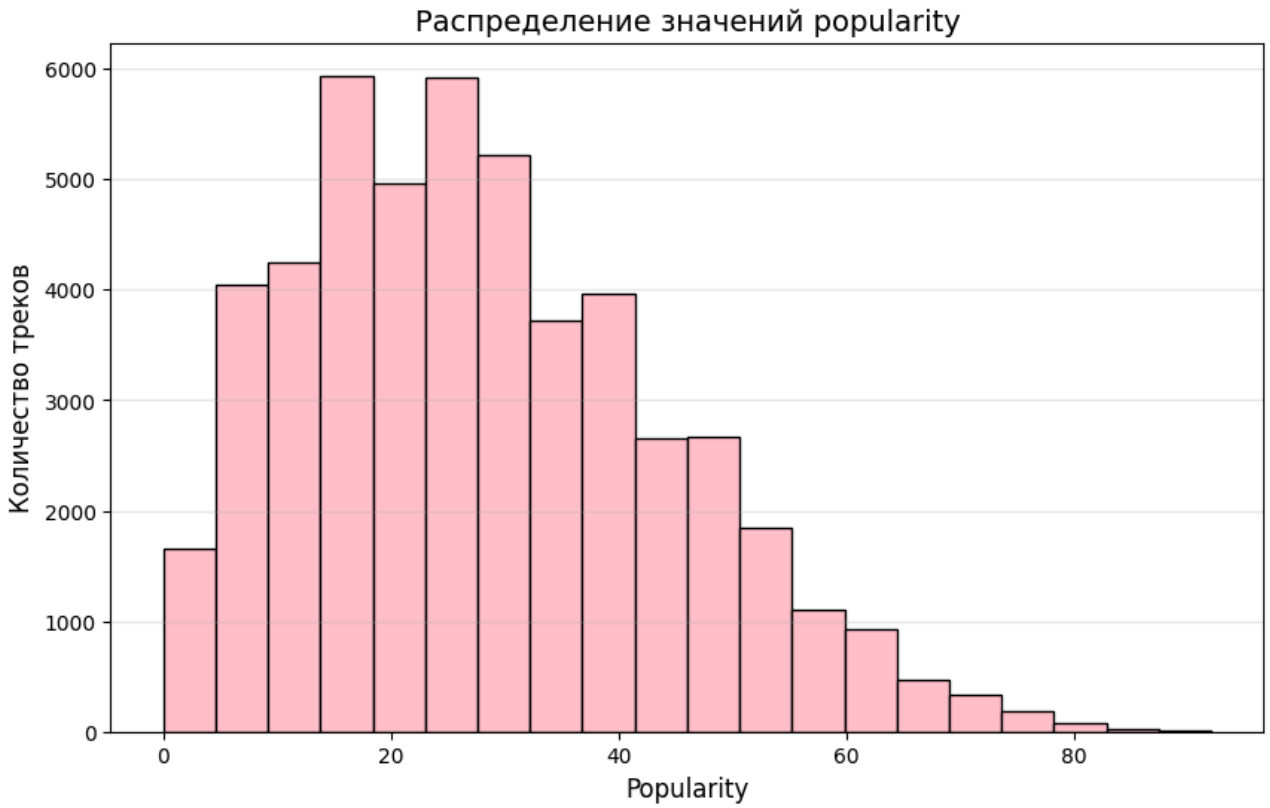
```
Мода для genre:
0    R&B
Name: genre, dtype: object
```

In [22…
```
mode_year = df['release_year'].mode()
print("Мода для release_year:")
print(mode_year)
```

```
Мода для release_year:
0    2024
Name: release_year, dtype: int64
```

Постройте гистограмму распределения значений popularity.

In [24…
```
plt.figure(figsize=(10, 6))
plt.hist(df['popularity'], bins=20, color='pink', edgecolor='black')
plt.title('Распределение значений popularity', fontsize=14)
plt.xlabel('Popularity', fontsize=12)
plt.ylabel('Количество треков', fontsize=12)
plt.grid(axis='y', alpha=0.3)
plt.show()
```


Распределение значений popularity
Количество треков
Popularity
0
1000
2000
3000
4000
5000
6000
0
20
40
60
80